SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA1703

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning,*
and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 50%

Duration: 1 Hour
QUESTIONS: 4 **Total: Marks:40**

Annexure A

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Code of Conduct:

I ***_D.VAISHNAVI(192524143)*** *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

(a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.

(b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).

(c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

1(a) Inductive Learning Approach

Inductive learning learns a general rule or model from a set of labelled examples and uses that model to predict unseen cases.

For this healthcare dataset:

- **Input features:** Age, Blood Pressure, Cholesterol, Glucose Level, BMI, Family History
- **Target variable: Diabetes** — Diabetic / Non-Diabetic

A suitable approach is **supervised inductive learning**, because the training records contain known diabetes outcomes.

Important features

Feature	Justification
Glucose Level	High glucose is strongly associated with diabetes and is an important predictor.
BMI	Higher BMI can be associated with increased diabetes risk.
Age	Diabetes risk generally changes with age.
Blood Pressure	Can provide additional information about the patient's health risk.
Cholesterol	Helps represent the patient's metabolic and cardiovascular condition.
Family History	Indicates inherited/genetic risk factors.

Conclusion: A supervised classification model can learn the relationship between these features and the diabetes class and predict whether a new patient is diabetic.

1(b) Handling Class Imbalance

The dataset contains **more non-diabetic cases than diabetic cases**, creating a class imbalance problem.

Suitable strategy: SMOTE

I would use **SMOTE (Synthetic Minority Over-sampling Technique)**.

SMOTE creates synthetic examples of the minority class rather than simply duplicating existing diabetic records.

Steps

1. Separate features and target.
2. Split the dataset into training and testing sets.
3. Apply **SMOTE only to the training data**.
4. Train the model using the balanced training set.
5. Evaluate using the original test set.

Why SMOTE?

It helps the model learn the characteristics of diabetic patients better and reduces the tendency to predict the majority class.

For a medical application, **Recall is particularly important**, because missing an actual diabetic patient is more serious than incorrectly identifying a non-diabetic patient as diabetic.

1c)Data Split and Preprocessing

Data split

Use:

- **70% → Training data**
- **30% → Testing data**

The 70:30 split provides enough data for the model to learn while retaining a sufficiently large independent test set for evaluation.

For medical data, the split should preferably be **stratified**, so that the diabetic/non-diabetic proportion is maintained in both sets.

Preprocessing

1. Normalisation

Continuous features such as:

- Age
- Blood Pressure
- Cholesterol
- Glucose
- BMI

can be scaled to a common range.

This is especially useful for models such as Logistic Regression.

2. Encoding

Categorical features such as **Family History** can be converted into numerical values.

Example:

- No → 0
- Yes → 1

Impact: Preprocessing improves consistency and allows machine-learning algorithms to process the data effectively.

Task 2: Decision Tree for Diagnosis

(a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.

(b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.

(c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

ANSWER:

2(a) Decision Tree

A Decision Tree classifier repeatedly divides the dataset based on feature values to reach a final prediction.

For example, the first three levels can be represented as:

```
              Glucose Level
             /          \
          High           Normal
          /                \
       BMI                  Age
      /   \               /    \
   High    Low          High     Low
    |       |             |      |
  Diabetic Non-         Diabetic Non-
           diabetic              diabetic
```

Note: The exact root and branches should be calculated from the actual dataset. The above is an appropriate illustrative structure based on the medical scenario; the uploaded question does not provide numerical patient records from which exact Information Gain/Gini values can be calculated.

Root-node selection

The root node is selected using Information Gain or Gini Index.

Information Gain:

IG=Entropy(parent)−Weighted Entropy(children)

The feature having the highest Information Gain is preferred.

For Gini Index:

Gini=1−∑pi2

The feature producing the lowest weighted Gini impurity is preferred.

For this healthcare problem, Glucose Level would be a reasonable candidate for the root because it is strongly related to diabetes, but the actual root must be determined from the dataset calculations.

2(b) Model Evaluation

The four important evaluation measures are:

Accuracy

Accuracy=TP+TN+FP+FNTP+TN

It measures the overall percentage of correct predictions.

Precision

Precision=TP+FPTP

It measures how many predicted diabetic cases are actually diabetic.

Recall

Recall=TP+FNTP

It measures how many actual diabetic patients are correctly detected.

F1-Score

F1=2×Precision+RecallPrecision×Recall

It provides a balance between precision and recall.

False Negative

A False Negative (FN) occurs when:

The patient is actually diabetic, but the model predicts non-diabetic.

This is particularly important in healthcare because a missed diabetes case may delay diagnosis and treatment.

How to reduce False Negatives

To reduce false negatives:

- Use SMOTE to balance the training data.
- Give greater importance to the minority class using class weights.
- Tune the classification threshold to improve recall.
- Evaluate using Recall and F1-score, not accuracy alone.

For clinical screening, a model with good recall/sensitivity is generally preferable when missing a disease case has serious consequences.

2(c) Pre-Pruning vs Post-Pruning

Pre-Pruning

The tree is restricted while it is being built.

Examples:

- Maximum depth
- Minimum samples per leaf
- Minimum samples for splitting

Post-Pruning

First build a larger tree and then remove branches that provide little predictive benefit.

The question specifically asks for cost-complexity pruning.

The objective can be represented as:

$R\alpha(T)=R(T)+\alpha|T|$

where:

- $R(T)$ = prediction error
- $|T|$ = number of terminal nodes
- α = complexity parameter

Comparison

Pre-Pruning	Post-Pruning
Stops tree growth early	Builds tree and then removes unnecessary branches
Faster	Can require more computation
May prevent overfitting	Usually provides better control of overfitting
May miss useful patterns	Can retain useful patterns before pruning

Clinical deployment

Post-pruning is generally preferable when it provides similar or better test performance with a simpler tree.

A smaller tree is easier for healthcare professionals to understand and explain, which is important for clinical decision-making.

However, the final choice should be based on test-set performance, recall, F1-score, AUC, and interpretability, not accuracy alone.

Task 3: Statistical Learning for Risk Stratification

(a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.

(b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

ANSWER:

3(a) Logistic Regression

A suitable statistical learning model is Logistic Regression because diabetes prediction is a binary classification problem.

The probability of diabetes can be represented as:

$P(Y=1)=1+e-z1$

where:

$z=\beta0+\beta1X1+\beta2X2+\cdots+\beta nXn$

Here:

- $Y=1 \rightarrow$ Diabetic
- $Y=0 \rightarrow$ Non-diabetic
- $Xi \rightarrow$ patient features
- $\beta i \rightarrow$ model coefficients

Comparison with Decision Tree

Metric	Decision Tree	Logistic Regression
Accuracy	Calculate from test data	Calculate from test data
AUC-ROC	Calculate from probabilities	Calculate from probabilities
Interpretability	High	High
Relationship	Can model nonlinear relationships	Mainly linear relationship in log-odds
Feature effects	Tree splits	Coefficients

Important: The uploaded assessment gives no actual patient dataset or numerical prediction results, so exact Accuracy and AUC-ROC values cannot be honestly calculated from the document alone.

When is Logistic Regression preferable?

Logistic Regression is preferable when:

- The relationship is reasonably linear.
- Interpretability is important.
- We need to understand how each feature affects risk.
- A relatively simple statistical model is desired.

3(b) Feature Importance

The assessment asks for the top three predictors from both the Decision Tree and statistical model.

For a Decision Tree, feature importance is obtained from the reduction in impurity produced by each feature.

For Logistic Regression, the magnitude of the coefficients can be examined after appropriate scaling.

Expected important predictors

Based on the healthcare scenario, likely important predictors include:

1. Glucose Level
2. BMI
3. Age / Family History

However, the exact top three must be calculated from the actual dataset. The document itself does not provide the data or calculated feature-importance values.

Do they need to agree?

Not necessarily.

Different algorithms measure feature importance differently:

- Decision Tree → reduction in impurity
- Logistic Regression → coefficient strength/effect on predicted log-odds

Therefore, some agreement is expected, but the rankings can differ.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.

(b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.

(c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

ANSWER:

4(a) Model as an MDP

The treatment recommendation problem can be formulated as a Markov Decision Process (MDP).

1. States

States represent the patient's health condition.

Example:

S0 = Low Risk
S1 = Moderate Risk
S2 = High Risk
S3 = Controlled/Improved

The state can be determined from factors such as glucose level, BMI, blood pressure and other health indicators.

2. Actions

The actions given in the question are:

A1 = Diet
A2 = Exercise
A3 = Medication
A4 = Monitor

3. Reward

The reward represents the health improvement obtained after an action.

Example:

Outcome	Reward
Significant improvement	+10
Moderate improvement	+5
No significant change	0
Health deterioration	-10

The actual reward values should be defined by the healthcare objectives and validated by medical professionals.

4. Transition Dynamics

The transition describes how the patient's state changes after an action.

Example:

High Risk + Exercise → Moderate Risk
Moderate Risk + Diet → Low Risk

High Risk + Monitor → High Risk
The probabilities of these transitions should ideally be learned from real longitudinal patient data.

4(b) Q-Learning
Q-Learning learns the value of taking an action in a particular state.
The update equation is:
$Q(s,a) \leftarrow Q(s,a)+\alpha[r+\gamma a' \max Q(s',a')-Q(s,a)]$
where:

- s = current state
- a = current action
- r = reward
- s′ = next state
- α = learning rate
- γ = discount factor

Example values

Assume:
$\alpha=0.5, \gamma=0.9$
Initial Q-values are 0.

Iteration 1

Suppose:

- Current state = High Risk
- Action = Exercise
- Reward = +5
- Next state = Moderate Risk
- Maximum next Q-value = 2

Then:
$Q(HighRisk,Exercise)=0+0.5[5+(0.9\times2)-0] =0.5[6.8] =3.4$

Iteration 2

Suppose the same state-action pair receives:

- Current Q = 3.4
- Reward = 10
- Maximum next Q-value = 4

Then:
$Q=3.4+0.5[10+(0.9\times4)-3.4] =3.4+0.5[10.2] =8.5$

Example Q-table

State	Diet	Exercise	Medication	Monitor
Low Risk	4.0	3.0	1.0	5.0
Moderate Risk	6.0	7.5	8.0	4.0
High Risk	5.0	8.5	9.0	3.0

This is an illustrative Q-table, because the assessment document does not provide actual patient transition data or rewards.

Five patient episodes

The agent should train for at least five episodes as required:

Episode	Starting State	Action	Reward	Next State
1	High Risk	Exercise	+5	Moderate
2	Moderate	Diet	+5	Low
3	High Risk	Medication	+10	Moderate

Episode	Starting State	Action	Reward	Next State
4	Moderate	Exercise	+5	Low
5	Low Risk	Monitor	+5	Low

Convergence criterion

Training can be considered converged when the change in Q-values becomes very small:

$\max|Q_{new} - Q_{old}| < \epsilon$

for a selected small threshold such as:

$\epsilon = 0.01$

over several consecutive episodes.

4(c) Final Learned Policy

The optimal policy is:

$\pi(s) = \arg\max Q(s,a)$

Using the illustrative Q-table above:

Patient State	Highest Q-value	Recommended Action
Low Risk	Monitor = 5.0	Monitor
Moderate Risk	Medication = 8.0	Medication
High Risk	Medication = 9.0	Medication

Important: This is only an example policy for demonstrating Q-Learning. It must not be treated as an actual medical treatment recommendation.

Ethical considerations

1. Patient Safety

An RL system should not independently make high-risk treatment decisions without qualified medical supervision.

2. Bias

If the training data is biased or unrepresentative, the model may produce unfair recommendations for certain patient groups.

3. Explainability

Healthcare professionals should be able to understand why a particular recommendation was generated.

4. Human Oversight

The final treatment decision should remain with a qualified healthcare professional.

5. Privacy

Patient information must be protected and handled according to applicable healthcare data-protection requirements.

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation